TRIGONOMETRIC INEQUALITIES

What is the first step?

Reduce the inequality to a basic one, then let the size of a decide its shape.

1 Reciprocal function: sec or csc?

YES → **Split by the sign of f(x)**
take the reciprocal carefully on each branch; asymptotes are always excluded → Reciprocal functions

NO ↓

2 Compound: $a \leqq f(x) \leqq b$?

YES → **Solve each side separately**
the solution is the intersection of the two sets
→ Compound inequalities

NO ↓

3 Transformed: $A \cdot f(Bx + C) + D$?

YES → **Isolate f, let u = Bx + C**
solve the basic form in u, then scale by $1/|B|$ and shift
→ Transformed functions

NO ↓

4 How large is $|a|$?
for sin or cos; tan takes every value

MORE THAN 1 → **All reals or empty**
no boundary equation needed
→ Sine inequalities

EXACTLY 1 → **Isolated points, all reals or empty**
compare a with the range [−1, 1]
→ Sine inequalities

LESS THAN 1 → **Solve f(x) = a for the boundaries**
a union of intervals, open or closed by strictness
→ Sine inequalities

THEN Repeat the pattern with the period: 2π for sin, cos, sec and csc; π for tan and cot.